Point decomposition beyond summation polynomials: literature review

and 14 leads **Date.** 2026-10-22 (dossiers completed 2026-10-23). **Type.** Literature search round with lead dossiers (research-visuals skill, "Deliverables for each search round"); no experiment was run. **Status of every statement.** *Proposed* unless a sentence says *derived* (desk algebra in a dossier, labelled unverified), *read at source* (a cited text was fetched and read), or *measured* (a number quoted from a frozen repository artefact, with its path). Nothing here changes a leaderboard, benchmark or scoreboard row, and no claim about a deployed curve is made or withdrawn. **Conductor.** Not installed in this container (the hooks warn and let edits through); the study touches only this directory.

Question. Which reformulations of the point decomposition problem (PDP) could replace or relax Semaev's summation polynomials in the relation search of elliptic-curve index calculus and lower the Gröbner-basis cost or the degree of regularity, measured against the current index-calculus baseline (`docs/ic/LEADERBOARD.md`: every row above matched rho, Koblitz 5.23×–21×, prime 5.47×–13.5×)?

Answer in one paragraph. The literature since 2020 is dormant on the summation-polynomial side (no new decomposition method; one 2026 preprint on the degree of regularity of descended S_3; learned Gröbner solvers and Crossbred theory on the solver side; two new generic-group lower bounds). Of the 14 leads drafted here, the director-tier dossiers close most of them: the MinRank, chord-lattice, trace-zero and prime-field block-structure reformulations are the same ideals or the same oracles the repository already has (LEAD-01, 06, 10, 14, each *closed-by-theorem* at the desk); the class filter and learned orderings are constant-factor levers that move no exponent and at most 1.1–2× at stage level (LEAD-04, 08); the McGuire–Mueller text and the elliptic-curve theorem of ePrint 2026/384 could not be read from this container (LEAD-11, 12, *unverifiable-from-here*); and the 2026 degree law is plausible for two-point S_3 but does not transfer to the chain routes (LEAD-05). The gate dossier (LEAD-12) finds no lead that the generic bounds exclude outright and three that could move the exponent in principle, which means "not excluded", not "indicated". The remaining five dossiers (02, 03, 07, 09, 13) are listed below with their status. The cheapest next measurement that decides several cards at once is LEAD-05's S_3 boundary replication (about 1 person-day, 10–15 CPU-hours), which LEAD-12 routes first.

Figures: `figures/pdp_landscape.svg` (source `figures/diagram.py`) and `figures/dreg_records.svg` (source `figures/plot_dreg.py`, data `figures/dreg_records.csv`). PDF: `report.pdf` (source `report.typ`). Literature map: `papers.csv`. Cards: `LEADS.md`. Dossiers and verification cards: `leads/`.

1. Scope and method

1. **Repository sweep** (2026-10-22, one read-only agent over `crypto`, `cryptanalysis`, `crypto-autoresearcher`, `ml-cryptanalysis`, `cryptanalysis-rl`): existing literature notes, code modules, ledger records, open problems, negative results and the baseline. The programme already holds a 25-file known-results map and about 60 literature notes on this topic in the autoresearcher's `knowledge/`, the master boundary ledger, two syntheses (`reports/Index calculus leads next steps.md`, `research/ic_leads_review_20261018/`) and native oracles for Semaev, Nagao, SAT/WDSat, root finding and Crossbred. The cards were written against that map so that closed ground is not re-opened (list in `LEADS.md`, "Standing constraints").
2. **External literature** (about 30 web searches and 12 page fetches, 2026-10-22): the summation-polynomial line from Semaev 2004 to Galbraith–Granger–Merz–Petit 2020, the degree-of-regularity line from Petit–Quisquater 2012 to Kousidis–Wiemers 2019 and the 2026 Solovyev preprint, the Gröbner-free and alternative-formulation line (Nagao, Faugère–Wallet summation ideals, McGuire–Mueller, Di Tullio–Pal, Gorla–Massierer trace zero), the solver line (WDSat, Crossbred, learned Gröbner bases) and the bound line (structured generic-group model, non-adaptive time–space bounds). `papers.csv` lists 44 sources with the access status each one had from this container: IACR ePrint PDFs are Cloudflare-blocked here, so most ePrint items are abstract-only.
3. **Lead cards** (`LEADS.md`): 14 cards, each with mechanism, why it could lower the Gröbner cost or the degree of regularity, prior art, the closed ground nearby, a discriminating test with a kill criterion, and repository hooks.
4. **Dispatch** (owner's model tiers, 2026-10-17): one director-tier Sonnet 5.5 agent per card wrote `leads/LEAD-NN.md` under `BRIEF_SONNET.md` (literature first, every citation marked read-at-source / abstract-only / recalled, no experiment, no research-state change); one mechanical-tier Haiku 5.5 agent per dossier wrote `leads/LEAD-NN.verify.md` under `BRIEF_HAIKU.md` (every citation re-fetched, every path tested with `test -e`, every quoted number grepped; computes nothing). The container restarted once during the round; the five dossiers that were lost (02, 03, 07, 09, 13) were re-dispatched on 2026-10-23 with a time budget.

2. What the literature says (summary of `papers.csv`)

- **The formulation is unchanged since 2004.** Every index calculus on elliptic curves over extension fields decomposes $Q = P_1 + \ldots + P_m$ by the zero set of Semaev's S_{m+1} (degree 2^{m-1} in each variable), Weil descent to n equations in $m \cdot \ell$ unknowns, symmetrisation under S_m (Faugère–Perret–Petit–Renault 2012) and under rational 2-torsion (Faugère–Huot–Joux–Renault–Vitse 2014), and a Gröbner or XL solve. The two published alternatives to S_{m+1} are Nagao's Riemann–Roch system ($(n^2-n)g$ quadrics; 2007/2010) and Faugère–Wallet's summation ideals on hyperelliptic Jacobians (2018), both for Jacobians on covers; the repository measured Nagao at prime n as "S'_4 with the linear part hidden"

(RESEARCH_PDP_SAT_SMT_NAGAO_SURVEY.md §5). McGuire–Mueller (ePrint 2017/1262) claim a Gröbner-free evaluation algorithm and a second one without summation polynomials; the text could not be read here (LEAD-11). Di Tullio–Pal (arXiv 1909.08901, read at source) write the prime-field DLP as quadratic conditions on functions y·g(x) + h(x) with a prescribed divisor, which is the Riemann–Roch form again.

- **The degree of regularity is the open quantity, and the literature is split on what to call it.** Petit–Quisquater's first fall m^2+1, Kousidis–Wiemers' m^2-m+1, Semaev's "≤ 4 under the first-fall-degree assumption", Huang–Kosters–Yeo's evidence against that assumption, and Kosters–Yeo's first fall 2 for descended S_3 all coexist. The only recent data point is Solovyev's 2026 preprint (GitHub release, no venue, single author): exact degree-capped closures giving S_3 degree 3, 4, 5 at $n \le 16$, $17–36$, $37–52$, then linear growth, with closed-form closure dimensions and S_4 solving degree = first fall = 7 up to $N = 24$. Figure 2 plots the recorded values side by side with the repository's own msolve (4) and XL (5–7) readings on the same cells.
- **Solvers moved, formulations did not.** WDSat (Trimoska–Ionica–Dequen 2019–2021) is measured here as an enumerator ($2^{3\ell}/3!$ nodes to within 0.2 %); Crossbred theory (Vidal–Ionica–Delaplace, ePrint 2024/992; preprocessing ePrint 2026/360) was never tried on the descent cells; learned Gröbner bases (Kera et al., NeurIPS 2024; HATSolver, arXiv 2512.14722) predict bases for 0-dimensional radical ideals and do not learn orderings (the LEAD-08 dossier corrects the card on this point).
- **Two 2025–2026 lower bounds fence the oracle space.** The structured generic-group model (Corrigan-Gibbs–Henzinger–Wu, ePrint 2026/384) and the non-adaptive time–space bounds (Dinur–Keller–Marmor, ePrint 2025/783) make every table-lookup or enumeration oracle a constant-factor lever; LEAD-12 maps each card onto them and records which statement is the paper's theorem and which is this repository's interpretation (the elliptic-curve theorem of 2026/384 was not readable from here).

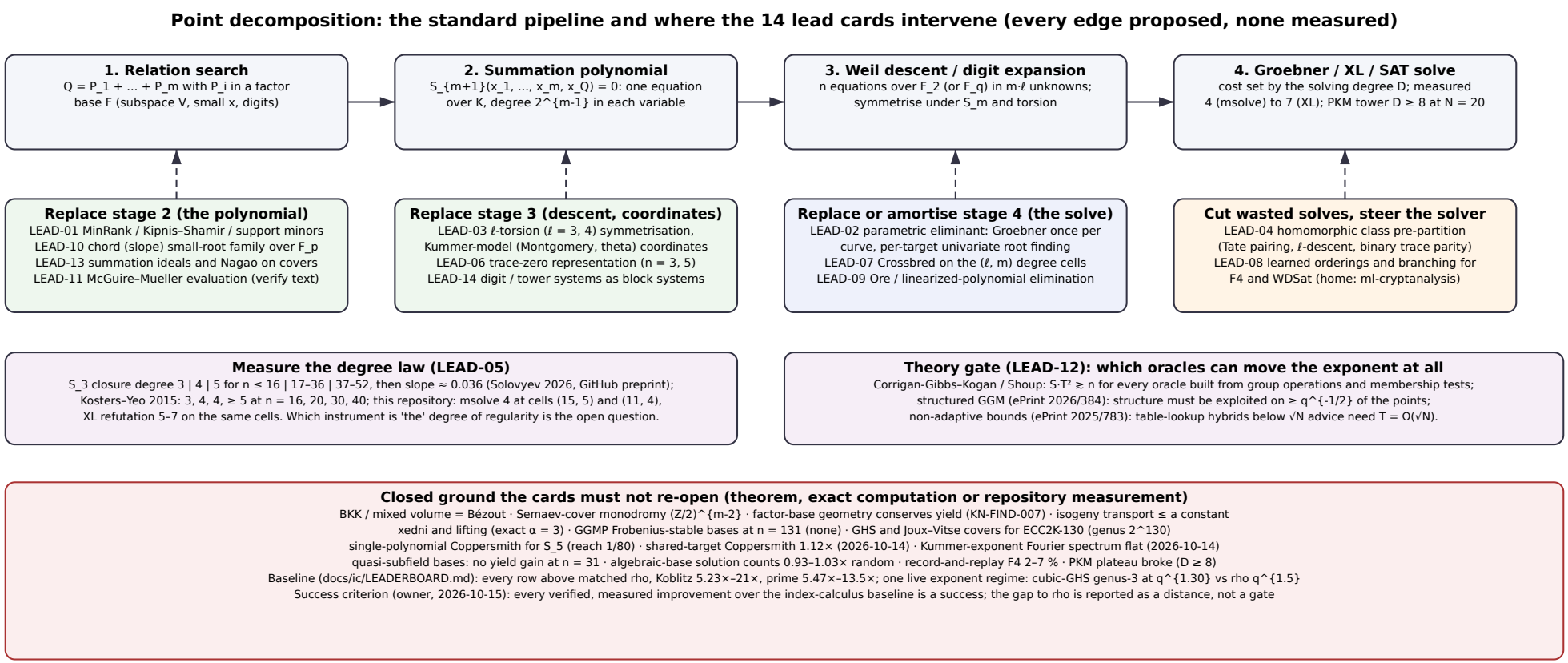


Figure 1. The standard PDP pipeline and where each lead card intervenes. Every edge is proposed; none is measured. Source: figures/diagram.py.

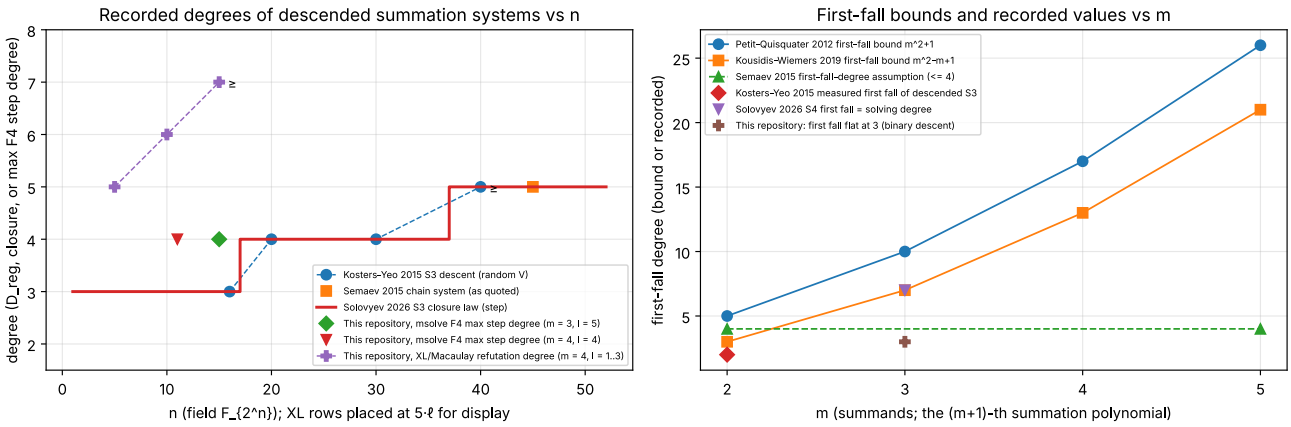


Figure 2. Recorded degree values for descended summation systems (left: degree versus n; right: first-fall bounds and recorded values versus m). Literature values are quoted as recorded, repository values name their file in figures/dreg_records.csv; no new computation. Source: figures/plot_dreg.py.

3. The 14 leads: verdicts

Verdicts are the dossiers' own (`leads/LEAD-NN.md`); "verification" is the mechanical card's totals (`leads/LEAD-NN.verify.md`: citations verified / mismatch / unreachable; paths exist / missing). Every mismatch and missing row was read by hand and is classified in §5; none is a wrong citation.

Lead	Card (class)	Dossier verdict
01	MinRank / Kipnis–Shamir / support-minors modelling of the summation chain (formulation)	**closed-by-theorem**: the Minors, KS and SM modellings are the repository's S_4-direct system, the chain, and the chain's ideal; none can read a lower degree than the chain. One residual probe (carry seeds) keeps about a 10 % prior of one degree
02	Parametric eliminant: generic-fibre Gröbner basis in the target's coordinates (solver, amortisation)	*pending (re-dispatched 2026-10-23)*
03	ℓ-torsion (3, 4) symmetrisation and Kummer-model coordinates (formulation)	*pending (re-dispatched 2026-10-23)*
04	Homomorphic class pre-partition: Tate pairing, ℓ-descent, trace parity (filter)	**constant-factor candidate**, cofactor curves only: `Z/r` has no proper quotient, so every prime-order row (all eight Table B prime rows) gets exactly 1.000×; pigeonhole caps the `m = 3` probe saving at 2× for any class group, so the card's "`|A|` in every oracle" is wrong
05	Degree-of-regularity growth law for descended S_3/S_4 (measurement)	**unverifiable-from-here**: the S_3 thresholds 3 ⟶ 4 ⟶ 5 at square `N = 16, 36` agree with Kosters–Yeo, Kousidis–Wiemers and the repository's own `lfd.sage` boundary, with one conflicting cell (Magma reads 4 at `n = 35` where the preprint reads 5); the law is for the quadratic two-point S_3, not the cubic-plus-quadratic chain, and the preprint's own S_4 data are flat at 7
06	Trace-zero variety representation for `E(F_{q^n})`, `n = 3, 5` (representation)	**closed-by-theorem**: `T_n` exists only for a curve defined over `F_q`, so it is not a coordinate system for the baseline's random prime-order `E/F_{p^k}`; at `n = 3` it ties rho at q^1; at `n = 5` the T_5 system is 9 equations of degree 32 in 8 unknowns against the `k = 4` ladder's 4 equations of degree ≤ 8
07	Crossbred on the Weil-descent degree cells (solver)	*pending (re-dispatched 2026-10-23)*
08	Learned orderings and branching for F4 and WDSat (solver, ML)	**constant-factor candidate**, stage level only: a variable order cannot change F4's solving degree on refuted or unique-solution targets (Caminata–Gorla Thm 3.1, Cor 3.3, read at source) and cannot lower WDSat's node count (thesis tables within +0.8 % of `2^{3ℓ}/3!`); only the F4 word-XOR cost is in play, where hand rules already give 2–4.91× at stage level with `S` unchanged
09	Ore / linearized-polynomial quotient-algebra elimination (solver)	*pending (re-dispatched 2026-10-23)*
10	Chord-parametrised small-root family over `F_p` (formulation)	**closed-by-theorem** for the small-unknown family: the chord system at `m = 2` is Semaev's S_3 in the symmetric (e_1, e_2) encoding after a free linear elimination of the slope (symbolic and numeric desk check); every encoding has the same limit reach `2/(m(m+1)d)` = 1/6, 1/24, 1/80

		and ceiling `1/(md)` = 1/4, 1/12, 1/32, so the `m = 4` kill fires (1/80 < 1/32 < 1/5)
11	McGuire–Mueller Gröbner-free algorithms, ePrint 2017/1262 (literature verification)	**unverifiable-from-here**: the PDF is Cloudflare-blocked (`cf-mitigated: challenge`), the Caen server times out, no mirror found; provisional classification from second-hand evidence: Algorithm 2 (sum `m`−1 base points, test membership) coincides with the repository's `rf-pairs`/`rf-mitm` oracles, Algorithm 1 is the same oracle with an algebraic evaluation step; both on the CGK line
12	Theory gate: structured GGM and non-adaptive bounds (theory)	**unverifiable-from-here** for the one statement it most wants (the elliptic-curve theorem of ePrint 2026/384 could not be read); the 14-row classification is delivered: 3 leads can move the exponent in principle (01, 06, 10 conditional), 9 constant only, 0 excluded outright, 2 are not oracles; none of CGK, Shoup, DKM or SGGM constrains an oracle that reads coordinates, so "in principle" means "not excluded"
13	Summation ideals (Faugère–Wallet) and Nagao on Joux–Vitse covers (formulation on the cover)	*pending (re-dispatched 2026-10-23)*
14	Prime-field digit and Kummer-tower systems as multi-homogeneous blocks (formulation)	**closed-by-theorem**: the bilinear and unmixed multilinear bounds do not apply (two structural hypotheses fail: the tower quadrics have block degree 2 and the system is over-determined); their formal values 7, 9, 11, 13 at `m = 3`, `N = 9`–`18` exceed the measured 6, 6, 7, 7 with a slope 3–7× too steep; any valid block bound has slope `(m−1)/m` in `N`, which the PKM note's §3.6 turns into at least √`r` per target

Table B. Ranking inputs and verification totals per lead.

Lead	Confidence	Expected gain if it works	Test cost	Verification (citations v/m/u; paths e/m)
01	medium	at most one degree (4 ⟶ 3): `m = 4` at `n = 131` moves +7.62 ⟶ +5.34 bits over rho (`w = 1`), still above	1–1.5 person-days, < 4 CPU-h, ≤ 1.5 GB (msolve)	12 / 2 / 0; 25 / 1
02				
03				
04	medium (bounds); low (effect on `S`)	about 1.1–1.2× on `S` for the cofactor-2/6 random-binary rows (desk estimate); exponent unchanged	1 person-day, < 1 CPU-h	2 / 0 / 0; 30 / 0
05	medium (`S₃` import); low (transfer)	none as an improvement; if the slope transferred, the `m = 3, 4` rows at `n = 131` would move *up*, to $2^{114.7-2}$129.1 and $2^{132.1-2}$155.6 column touches	1 person-day, 10–15 CPU-h, ≤ 12 GB	8 / 0 / 0; 24 / 1
06	high (no improvement); medium (`n = 5` asymptotics)	none against any baseline row; a `T_n` row would be a new baseline, not an improvement	2–3 person-days (new `T₃` harness), ≤ 2 CPU-h	1 / 4 / 4; 16 / 0
07				
08	medium	0 degree levels; WDSat ≤ 1.04×; F4 XORs 1.0–2× beyond the hand rules (guess); `S` 1.00×	2 person-days core, 10 CPU-h mechanical, ≤ 16 GB	7 / 0 / 5; 21 / 3
09				
10	medium	none in the exponent; 5–11 % higher `δ` at equal multiplicity (2–3× smaller lattices) on oracles that cannot beat rho	zero-compute test done in the dossier; 0.5 person-day to confirm in the `05` MILP harness	6 / 0 / 1; 12 / 0
11	low (text); medium (classification)	none; the relayed `O(p)` cost is a worse exponent than Table B's `m = 2` row (`r^{2/3}`)	0.5 person-day once a host can download the CC BY PDF	9 / 2 / 3; 13 / 0
12	medium (generic rows); low (algebraic rows)	none (a gate); routes the first test to LEAD-05	0.5 person-day to close the reading gap	8 / 1 / 3; 10 / 0
13				

14	medium	none on the exponent, and the sign is wrong; constant unknown, likely ≤ 1.5×	0.5 person-day, < 1 CPU-minute	10 / 0 / 0; 12 / 1

3.1 What the completed dossiers add beyond their verdicts

- **LEAD-01.** The Sylvester determinant of the $S_3 \circ S_3$ chain is S_4 (desk check); Bardet–Bertin Prop. 1 (read at source) gives KS = SM ideals with Minors ⊂ KS, so the three modellings are generating sets of ideals the repository already solves. The bilinear bound does not apply because the chain has the cubic term x_1x_2y; read naively it gives 16 at `(15, 5)` against a measured F4 degree of 4. The free-kernel KS system is 14.7–16.4× wider at degree 4. The one thing left is whether the closed-form carry seeds (bit components of x_1f_1, x_2f_1, yf_1) lie in the degree-4 Macaulay span and let F4 finish at 3; the dossier replaces the card's test with that probe (R1).
- **LEAD-04.** Why KN-FIND-14706d does not cover this: that finding tests the product character $\chi_1\chi_2\chi_3$, which is trivial on `E(F_p)` by the curve equation; the 2-descent class (χ_1, χ_2) lives in its kernel and takes all four values. But on a prime-order row there are no classes at all, and on cofactor rows hash lookups and membership tests are already exact and uncharged, so only table-build additions fall. The dossier flags an adjacent lever outside the card: the random-binary ladder builds its base from raw points with no projection into the subgroup; a class-0 or cofactor-projected base is estimated at 1.4–1.6× at fixed base size on `f2m21` (desk estimate, no record of it being tried).
- **LEAD-05.** Solovyev's "degree of regularity" is the degree-capped closure with iterated falls (Huang–Kosters–Yeo Def. 1), which matches the repository's MutantXL closure and msolve's F4 step degree draw for draw and sits below XL refutation. The M4GB fix is PR #14 of `cr-marcstevens/m4gb`, opened by the preprint's author on 18 Sep 2026 and merged 25 Sep 2026. The preprint's "5 for 37 ≤ n ≤ 52" hides a parity split (odd `n = 51` predicted 6). The decisive cells beyond the cheap test (S_4 at `N = 27`, S_3 at `N ≈ 52`, chain `m = 4`, ℓ `= 6`) need 37–128 GB hosts.
- **LEAD-06.** The prime-order subgroup lies in `T_n` iff `gcd(ℓ, #E(F_q)) = 1`, automatic for $\ell > q + 1 + 2\sqrt{q}$ but capped at $\ell \le$ `#T_n ≈ q^{n−1}`. The one positive is a cheaper per-residual solve (2 unknowns, 6 solutions, yield 1/2 against 3 unknowns, 64 solutions, yield 1/6 for S_4), which comes from the smaller group and does not transfer to generic curves. The card's pointer "ePrint 2014/158" is the point-compression paper; the optimal-representation paper is arXiv 1405.2733 / ePrint 2014/333.
- **LEAD-08.** HATSolver predicts the lex Gröbner basis directly rather than learning an ordering; the relevant learned-order paper is Bunch et al., arXiv 2602.02972. The instance-level degree is a cell-level constant, identical across targets in the committed data, so there is nothing per-instance to learn; the test is replaced by a held-out-cell split with a permutation null, runnable from ml-cryptanalysis.
- **LEAD-10.** Two side findings: `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §4.6 puts the (e_1, e_2) reach at 1/9, which W10's own e-encoding rows already exceed (the family value is 1/6); W10's `m = 3` e-encoding rows were run in a box that binds, so 0.018176 and 0.024903 are lower bounds (a non-binding desk LP gives 0.0201 and 0.0269, unconfirmed in W10's Rust engine). The symmetric encoding also enlarges the root set: about `4p^{1/2}` slopes with small s_1 per genuine decomposition at `B = p^{1/2}`.
- **LEAD-11.** Corrections for `KN-LIT-6966ea` are listed in the dossier's §8; the "McGuire–Mueller trinomial bound" cited in `research/ic_next10_20261014/08-quasi-subfield/README.md:53` is their 2019 arXiv paper (1905.11755, read at source), not 2017/1262.
- **LEAD-12.** The $S \cdot T^2 \gtrsim N/|\mathrm{Aut}|$ form used across the repository is in none of the four bound papers; unique factorisation fails for elliptic-curve sums, so every elliptic-curve use of δ is the repository's derivation. The most uncertain classification is LEAD-03, with 01, 07 and 14 riding on it: a constant-factor drop in the slope of `D(n)` would move the exponent (desk check: cost exponent per bit of `n` 0.81 at slope 1/20 and 0.48 at 1/40, against rho's 0.5), which is why LEAD-05's fit of `D(n)` is the first test.
- **LEAD-14.** The card's premise is wrong at source: Petit–Kosters–Messeng use a composition tower, not binary digits, and their experiments are `m = 2` only. Their Tables 4–5 (Magma, `m = 2`) independently reproduce the repository's `D = 4, 5, 6` rises, which `RESEARCH_PKM_TOWER_ORACLE.md` §10.10 had recorded as unchecked. Kudo et al. is located as CANS 2018, LNCS 11124.

4. Ranking and what to do next

(Provisional until the five re-dispatched dossiers land; the ranking below uses the nine completed ones and LEAD-12's gate.)

1. **Run LEAD-05's cheap test first** (about 1 person-day, 10–15 CPU-hours, ≤ 12 GB): replicate the S_3 closure boundaries at `N = 16/17` and `36/37` with the repository's closure driver and msolve, with `n = 35` as the decisive cell against Kousidis–Wiemers' Magma reading, and check three closure-dimension formula points. It decides whether the 2026 law can be imported at all, and LEAD-12 makes it the gate for 01, 03, 07 and 14.
2. **LEAD-01's residual probe R1** (1–1.5 person-days, < 4 CPU-hours): export the closed-form carry seeds and test whether msolve finishes at degree 3 on the `(15, 5)` and `(11, 4)` cells. About a 10 % prior; the prize is one degree, worth 7–65× in Macaulay width.

3. **LEAD-04's arm C** (1 person-day): the cofactor-projected base on the random-binary ladder, 1.4–1.6× estimated at fixed base size, with the prime-order negative control that must read 1.000×. A measured improvement over a leaderboard row would count under the success criterion even at that size.
4. **Do not spend on** LEAD-06, 10, 14 (closed at the desk, with the reasoning in the dossiers) or LEAD-08 beyond a cheap null test (the theorem leaves only matrix cost, where hand rules already exist).
5. **Reading debt** (0.5 person-day each, needs a host outside the Cloudflare block): ePrint 2017/1262 (LEAD-11) and the elliptic-curve section of ePrint 2026/384 (LEAD-12). Until then both verdicts stay *unverifiable-from-here*.

The gap to matched rho is unchanged by this round: no row moves, and the distance remains what `docs/ic/LEADERBOARD.md` records.

5. Verification (mechanical tier)

Each `leads/LEAD-NN.verify.md` re-fetched every citation, tested every path and grepped every quoted number, recording outcomes verbatim. The non-clean rows, read by hand:

- **Citation mismatches.** All are year-of-version or wording differences: arXiv submission year against journal year (LEAD-06 ×3, LEAD-12 ×1 where a search ranked a different paper first and the right one second), title capitalisation or an abbreviated title in the dossier (LEAD-01 ×2, LEAD-11 ×2), and a thesis whose host (`edoc-vmtest.ub.unibas.ch`) did not resolve (LEAD-06). No row shows a different paper or different authors.
- **Unreachable citations.** ePrint PDFs and some abstract pages (403 or a Cloudflare challenge), one Springer chapter behind a login, the Caen 2018 server (504), Semantic Scholar's rate limit. The dossiers mark these abstract-only or recalled.
- **Missing paths.** Files the discriminating tests propose to create (`examples/lead01_carry_seeds.rs`, `examples/lead14_block_bounds.rs`, `examples/pdp_perm_f4.rs`, `csd/order_probe.py`), one external repository file (`cr-marcstevens/m4gb src/m4gb.hpp`), and one brace-expansion artefact (`README` for `README.md`). No cited existing path is missing.
- **Numbers not found by literal grep.** Formatting differences ("0 of 345" for "0/345", "8 of 8", "N = 40 and 42"; confirmed by hand in the named files), and in LEAD-14 three numbers that live in `research/pkm_m3_n21_msolve_20261018/RESULTS.md` while the dossier names the sibling `README.md` (an attribution slip, not a wrong number). One LEAD-05 number ("37 of 38" in `fall_degree_bounds_20261008/RESULTS.md`) was not located by hand either and stays unconfirmed.

6. Corrections to repository records surfaced by this round

Recorded here for the owners of those files; this PR changes none of them.

- `knowledge/literature/KN-LIT-6966ea.md` (autoresearcher) is abstract-only and should carry the corrections in `leads/LEAD-11.md` §8.
- `research/ic_next10_20261014/08-quasi-subfield/README.md:53` cites a "McGuire–Mueller trinomial bound" to ePrint 2017/1262; the source is arXiv 1905.11755.
- `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §4.6's (e_1, e_2) reach of 1/9 is a basic-strategy figure; the family value is 1/6 (LEAD-10).
- `RESEARCH_PKM_TOWER_ORACLE.md` §10.10's "unchecked" against Petit–Kosters–Messeng: their Tables 4–5 reproduce the $D = 4, 5, 6$ rises (LEAD-14).
- `LEADS.md` card errors found by the dossiers and left in place for the record: the Solovyev preprint is 2026, not 2025 (corrected in the card heading and figures); "ePrint 2014/158" is the point-compression paper (LEAD-06); HATSolver does not learn orderings (LEAD-08); PKM use a composition tower, not binary digits (LEAD-14); "$|A|$ in every oracle" overstates the pair-table saving (LEAD-04).

7. Unresolved checks and access failures

- IACR ePrint PDFs are blocked from this container by a Cloudflare challenge; WebFetch on the same URLs returns 403. The texts that matter most and were not read: ePrint 2017/1262 (LEAD-11), the elliptic-curve section of ePrint 2026/384 (LEAD-12), ePrint 2015/573 (LEAD-05), ePrint 2019/313 (LEAD-08; covered by Trimoska's thesis, read at source).
- The Caen 2018 talk and paper by Mueller (nitaj.users.lmno.cnrs.fr) returned 504; Mueller's UCD thesis was not found online.
- The five re-dispatched dossiers (02, 03, 07, 09, 13) are listed as pending in §3 until their agents report; this README and the PDF are regenerated when they do.

8. Graphs checked and left unchanged

No measured result was produced, so no canonical graph changes: `docs/index-calculus-scoreboard.html`, `docs/ic/progress-timeline.json`, `docs/ic-leaderboard.html` / `docs/ic/LEADERBOARD.md` / `docs/ic/leaderboard.json`, `docs/performance-gains/`, and `benchmarks/README.md` are untouched. Nothing here is a performance result, so no `benchmarks/` entry is owed.

9. Files

File	What it is

`LEADS.md`	The 14 lead cards and the standing constraints
`BRIEF_SONNET.md`, `BRIEF_HAIKU.md`	The director-tier and mechanical-tier briefs
`leads/LEAD-NN.md`	Director-tier dossiers (Sonnet 5.5), one per card
`leads/LEAD-NN.verify.md`	Mechanical-tier verification cards (Haiku 5.5), one per dossier
`papers.csv`	Literature map, 44 sources, access status per source
`figures/pdp_landscape.svg`, `.png`, `diagram.py`	Figure 1 and its source
`figures/dreg_records.svg`, `.png`, `.csv`, `plot_dreg.py`	Figure 2, its data and its source
`report.typ`, `report.pdf`	This report rendered (pandoc → Typst)